DATASHEET

8Bit_Counter_Integrated

8-Bit Synchronous Counter with Serial Output

Technology	Supply	Clock	Library
180nm CMOS	**1.8 V (VDD)**	**50 MHz**	**8BitCounter_PHY_ED222**

Group – 4

Members

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The **8Bit_Counter_Integrated** is a full-custom, transistor-level 8-bit synchronous up-counter with a pin-constrained serial output architecture. Implemented entirely from NAND gates in the 180nm CMOS process, the design integrates an 8-bit synchronous counter with a parallel-in serial-out (PISO) shift register, delivering the complete 8-bit count value as a serial bit stream on a single output pin (SOUT). This architecture fits within a total of 6 functional I/O pins, making it suitable for highly pin-constrained chip integration scenarios.

DESIGN MODIFICATIONS vs. standard 8-bit counter:

1. Serial output (PISO) replaces 8 parallel outputs — 1 pin vs 8 pins
2. Pin-constrained architecture: 6 functional pins total (+ VDD + GND)
3. Full custom transistor-level design

Parameter	Value
Top module	8Bit_Counter_Integrated
Course	ED222 — Digital IC Design and Tape out Lab
Main library	8BitCounter_PHY_ED222
Gates library	Gates_ED222_PY
Tool	Cadence Virtuoso
Design style	Full custom — transistor level, NAND-based
Base technology	180nm CMOS
Nominal VDD	1.8 V

1 Pin Configuration

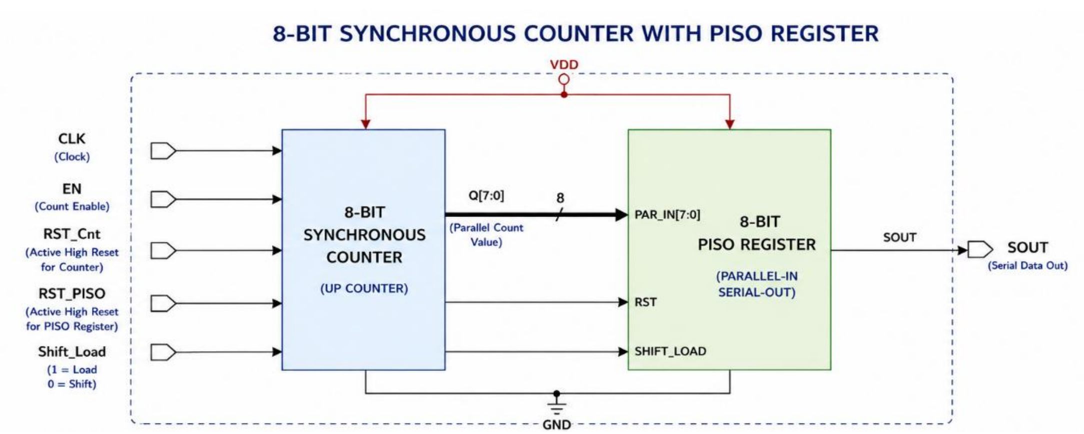
The top-level module **8Bit_Counter_Integrated** exposes 6 functional signal pins plus VDD and GND.

Pin	Direction	Active Level	Description
CLK	Input	Positive edge	Master clock — shared by both the synchronous counter and the PISO shift register
RST_CNT	Input	Active HIGH	Synchronous reset for the 8-bit counter only; clears Q[7:0] to 0x00 on the next rising CLK edge
RST_PISO	Input	Active HIGH	Synchronous reset for the PISO shift register only; clears all stages to 0
EN	Input	Active HIGH	Counter enable — counter increments only when EN = 1 for exactly 1 CLK cycle; must not overlap with SHIFT_LOAD = 0
SHIFT_LOAD	Input	0 = Load 1 = Shift	PISO mode select: 0 fires a 1-cycle parallel load from counter Q[7:0]; 1 enables serial shifting
SOUT	Output	Active HIGH	Serial data output — MSB first; Q[7] appears first, Q[0] last; one bit per CLK cycle during shift phase
VDD	Power	—	Supply voltage — nominal 1.8 V
GND	Power	—	Ground reference — 0 V

> Note — Dual reset design (modification): RST_CNT and RST_PISO are independent, allowing the counter and PISO shift register to be reset separately. This is a deliberate departure from a standard single-RST counter, enabling finer control during initialisation and test.

2 Block Diagram

The diagram below shows the top-level signal flow. The counter and PISO are both driven by a single **CLK** rail. The 8-bit parallel bus Q[7:0] is an internal signal only — it never appears on a pin.



2.1 8-Bit Synchronous Counter

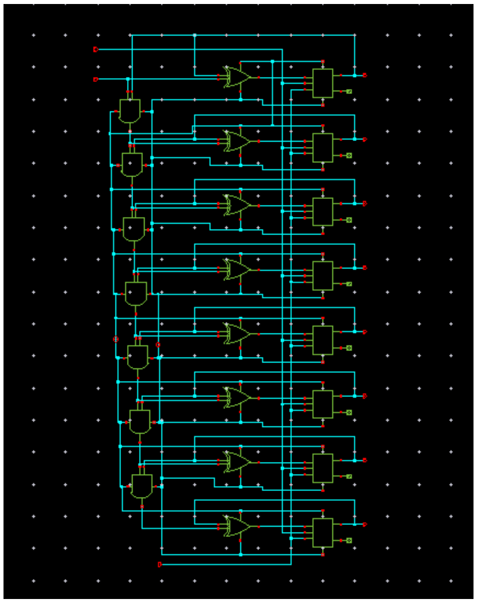


Figure 1: Schematic

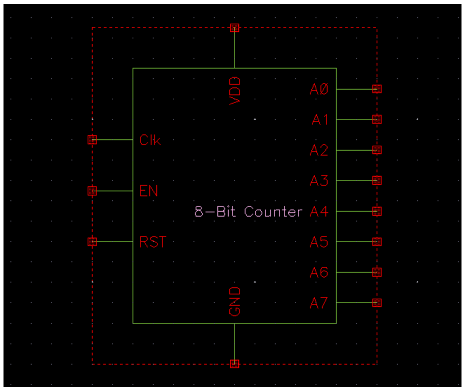


Figure 2: Symbol

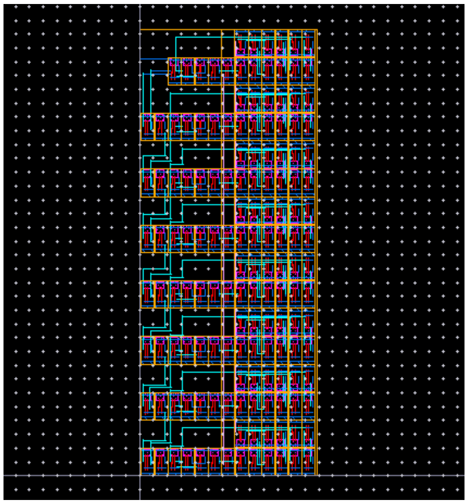
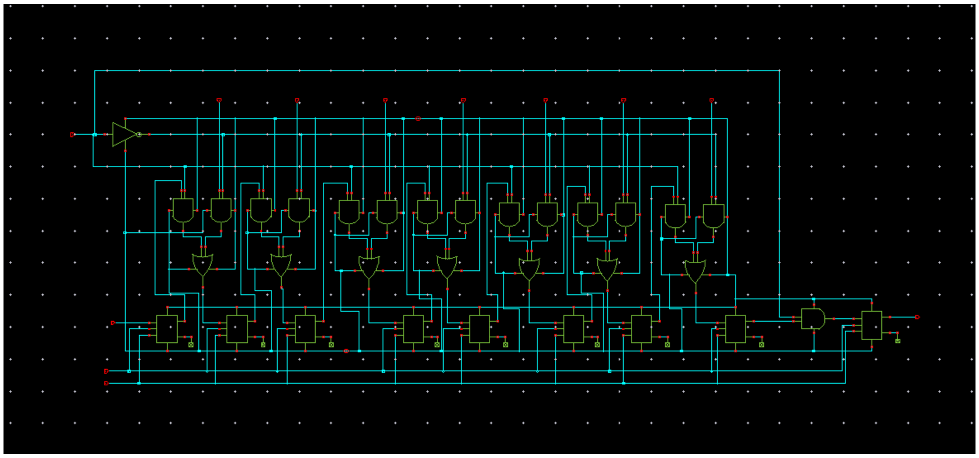
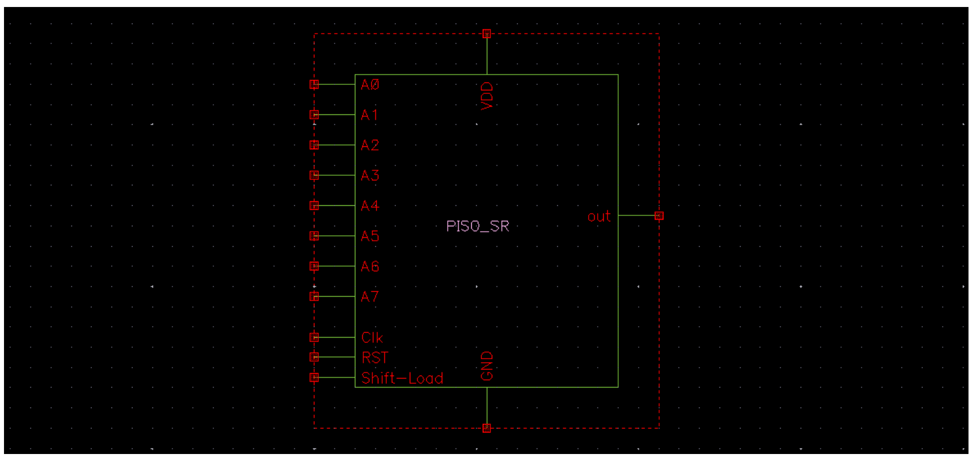
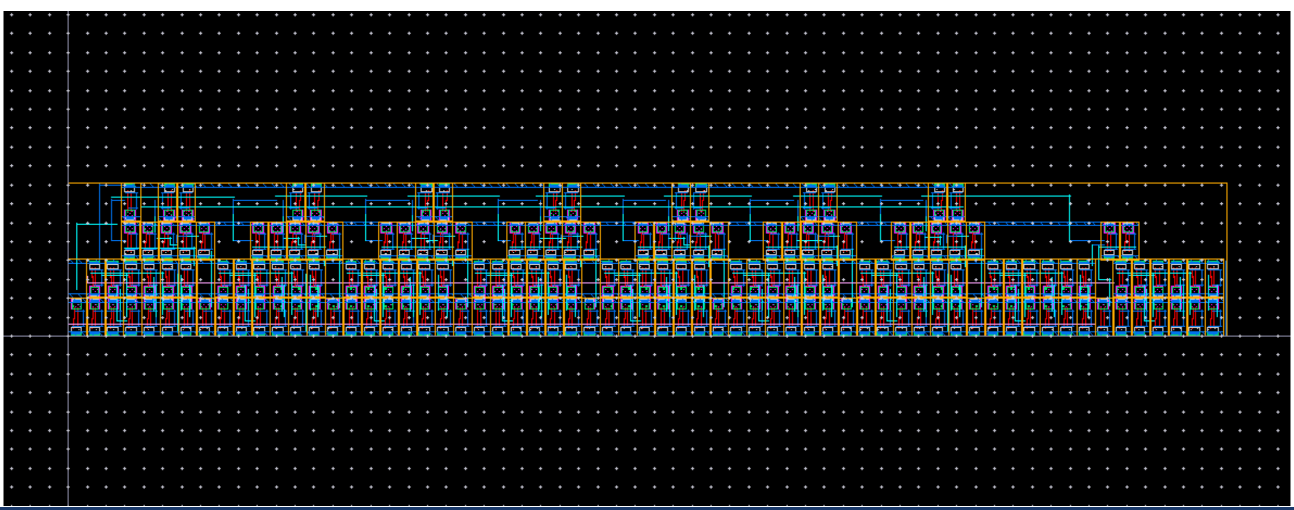


Figure 3: Layout

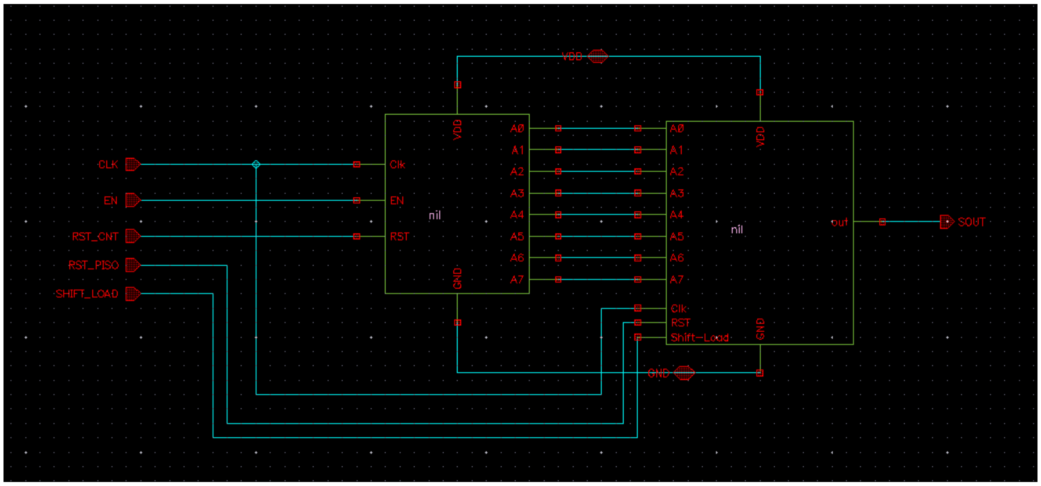
2.2 PISO

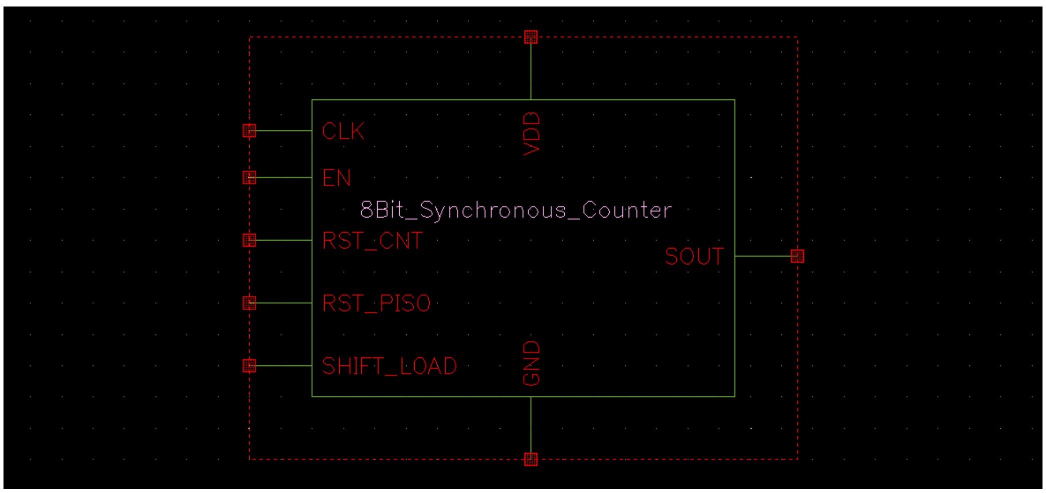


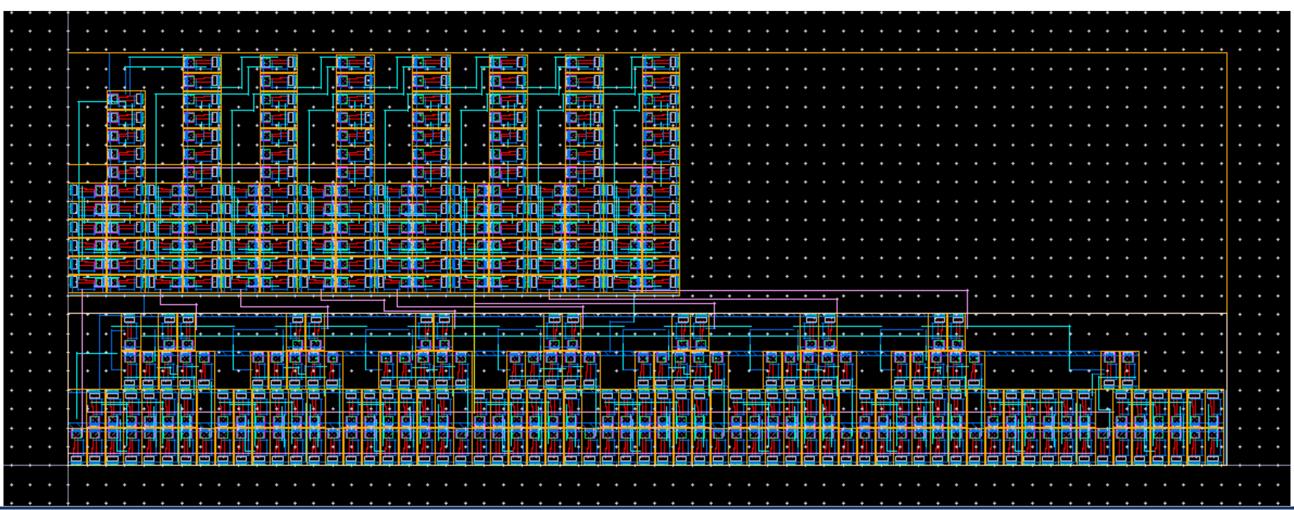


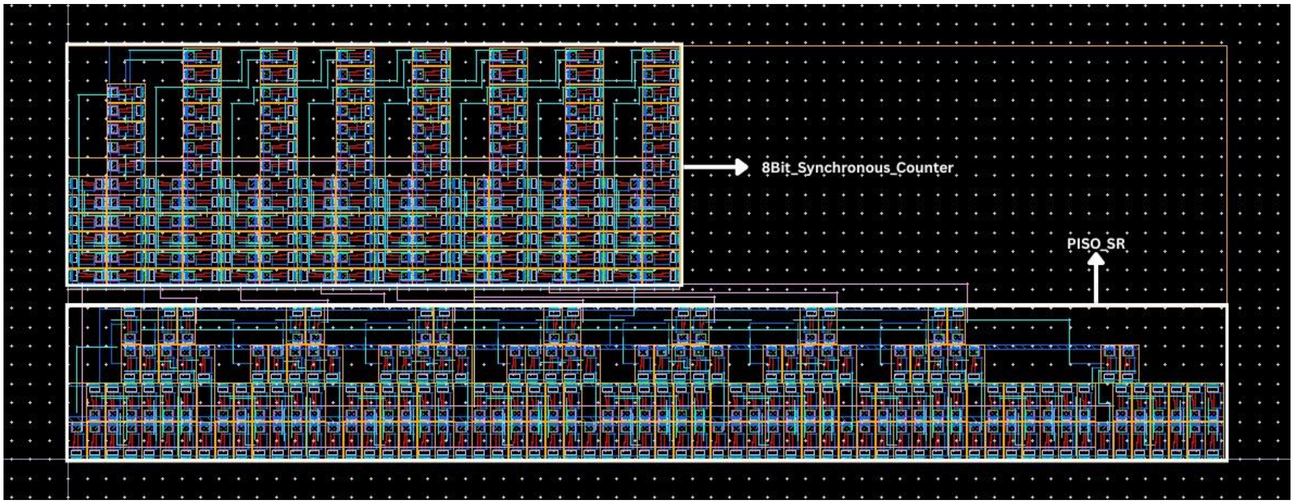


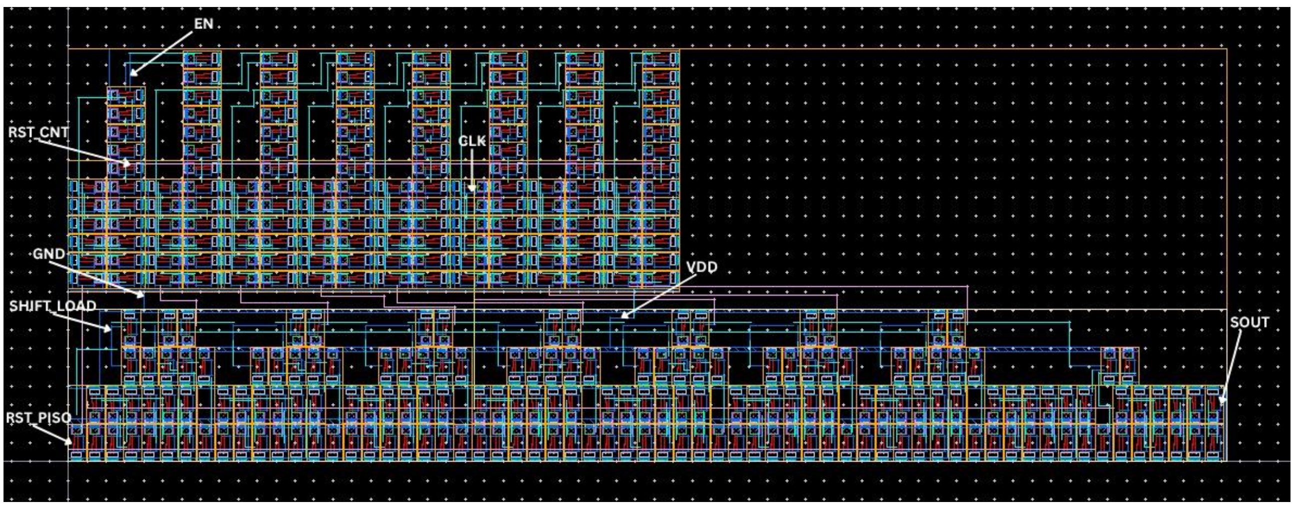
2.3 Counter + PISO











2.5 Layout Area Analysis

Width = 306.415 um
Height = 109.385 um

$$\text{Area} = \text{Width} \times \text{Height}$$

$$\text{Area} = \mathbf{306.415\ \mu m} \times \mathbf{109.385\ \mu m}$$

$$\text{Area} = \mathbf{33515.507275\ \mu m^2}$$

3 Functional Description

3.1 8-Bit Synchronous Counter

The counter is a synchronous binary up-counter — all 8 D flip-flops share a single **CLK** edge. The carry chain is implemented using **AND_with_NAND** and **EXOR_with_NAND** cells. The counter increments on every rising CLK edge when **EN = 1**. When EN = 0, the count is held (frozen), allowing the PISO to read a stable value.

Condition	Counter behaviour
RST_CNT = 1, CLK ↑	Counter resets to 0x00 synchronously, overrides EN
RST_CNT = 0, EN = 1, CLK ↑	Counter increments by 1 (Q[7:0] ← Q[7:0] + 1)
RST_CNT = 0, EN = 0	Counter holds current value — no change regardless of CLK
Overflow (0xFF → 0x00)	Wraps to 0x00 automatically on the next increment

3.2 PISO Shift Register

The PISO shift register captures the 8-bit counter value in a single CLK cycle **(when SHIFT_LOAD = 0)** and then shifts it out serially over 8 CLK cycles **(SHIFT_LOAD = 1)**. Output is MSB-first (Q[7] first, Q[0] last).

SHIFT_LOAD	PISO action
0 (Load)	On rising CLK edge: D[7:0] ← Counter Q[7:0] (parallel capture in one cycle)
1 (Shift)	On each rising CLK edge: stages shift right; MSB exits through SOUT

4 Operating Sequence — 10-Cycle Frame

One complete output frame consists of exactly 10 clock cycles (200 ns at 50 MHz). The three phases must be executed in order. **EN and SHIFT_LOAD = 0 must never assert on the same CLK edge.**

#	Phase	Signal Conditions & Action	Duration	Time @ 50 MHz
1	**Count**	EN = 1; SHIFT_LOAD = 1 (or don't care) Counter increments by 1 on the rising CLK edge.	1 cycle	20 ns
2	**Load**	EN = 0; SHIFT_LOAD = 0 PISO captures counter Q[7:0] in parallel on the rising CLK edge.	1 cycle	20 ns
3	**Shift × 8**	EN = 0; SHIFT_LOAD = 1 PISO shifts one bit per CLK. SOUT outputs Q[7] → Q[6] → … → Q[0].	8 cycles	160 ns

Frame rate: 50 MHz ÷ 10 = 5 MHz (one complete 8-bit value per 200 ns)
Serial data rate: 50 Mbit/s (during the 8-cycle shift phase)

4.1 Critical Timing Constraints

Constraint	Rule	Consequence of violation
EN ↔ SHIFT_LOAD separation	EN = 1 on cycle C1; SHIFT_LOAD = 0 on cycle C2 (the very next cycle). Never on the same edge.	Counter increments and PISO loads simultaneously — PISO captures the old (pre-increment) value, producing an off-by-one error in SOUT
SHIFT_LOAD = 0 pulse width	Exactly 1 CLK cycle	Holding low for >1 cycle re-loads the (now incremented) counter value, corrupting the frame
SHIFT_LOAD = 1 duration	Exactly 8 CLK cycles	Fewer cycles → incomplete readout; more cycles → zeros shifted in from empty stages
RST_CNT / RST_PISO hold	Must de-assert before the cycle where EN or SHIFT_LOAD are asserted	Reset and load/count overlap may produce metastability in the DFF master-slave latch

5 Timing Diagram (One Frame — 10 CLK cycles)

The table below shows signal states across one complete frame. CLK period = 20 ns (PW = 10 ns). N denotes the counter value at the start of the frame.

Signal	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
CLK ↑ edge	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
EN	**1**	0	0	0	0	0	0	0	0	0
SHIFT_LOAD	—	**0 (LOAD)**	1 (SHIFT)	1 (SHIFT)	1 (SHIFT)	1 (SHIFT)	1 (SHIFT)	1 (SHIFT)	1 (SHIFT)	1 (SHIFT)
RST_CNT	0	0	0	0	0	0	0	0	0	0
RST_PISO	0	0	0	0	0	0	0	0	0	0
Counter	**N→N+1**	N+1 (frozen)	N+1 (frozen)	N+1 (frozen)	N+1 (frozen)	N+1 (frozen)	N+1 (frozen)	N+1 (frozen)	N+1 (frozen)	N+1 (frozen)
SOUT	—	—	**Q[7]**	**Q[6]**	**Q[5]**	**Q[4]**	**Q[3]**	**Q[2]**	**Q[1]**	**Q[0]**

Total Clock Cycles required: 10 CLK cycles x 256 (0 -> 255) = 2560 Cycles
Total Simulation Time, t_{sim} required: 10 CLK cycles x T_{CLK} x 256 (0 -> 255) = 51200 ns
SOUT bit order during shift phase (C3–C10):

C3 = Q[7] (MSB)   C7 = Q[3]
C4 = Q[6]   C8 = Q[2]
C5 = Q[5]   C9 = Q[1]

C6 = Q[4]   C10 = Q[0] (LSB)

5.1 Timing Analysis (Functional Verification)

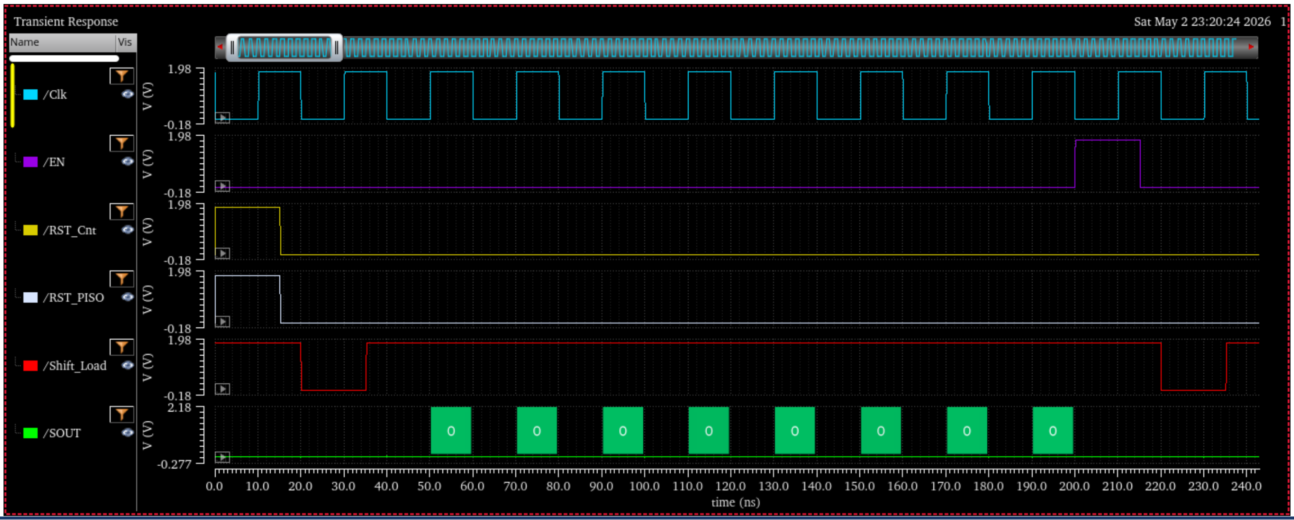


Figure 4: 0x00

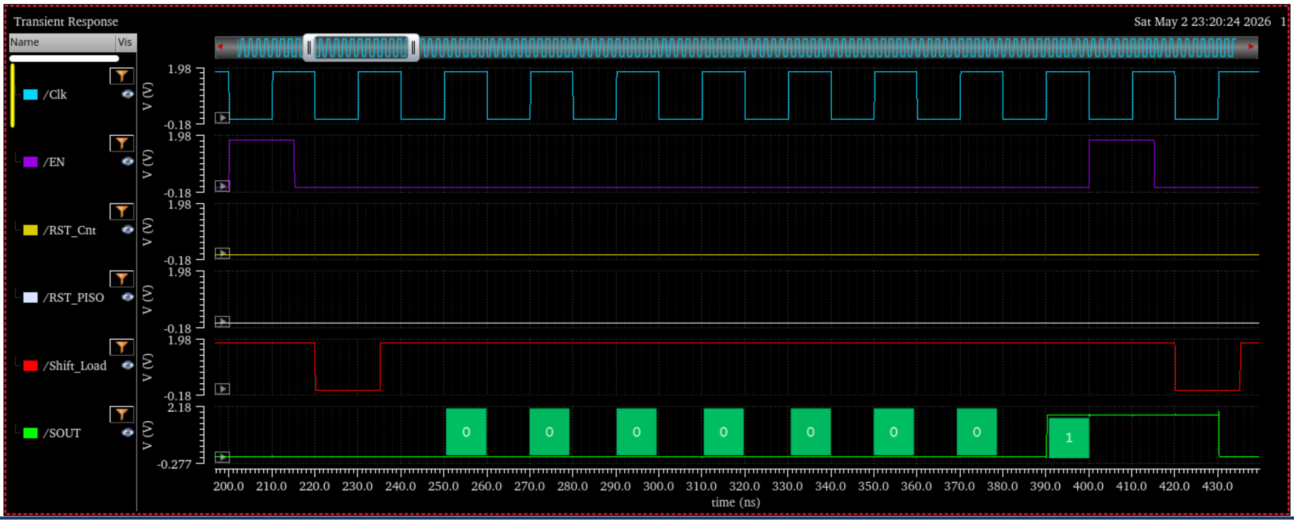


Figure 5: 0x01

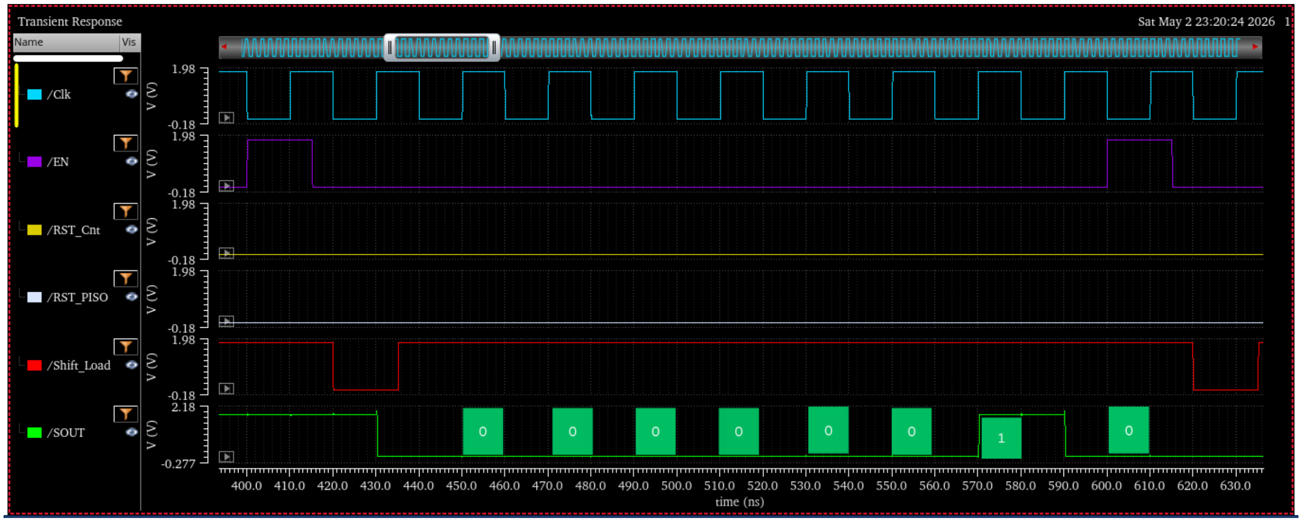


Figure 6: 0x02

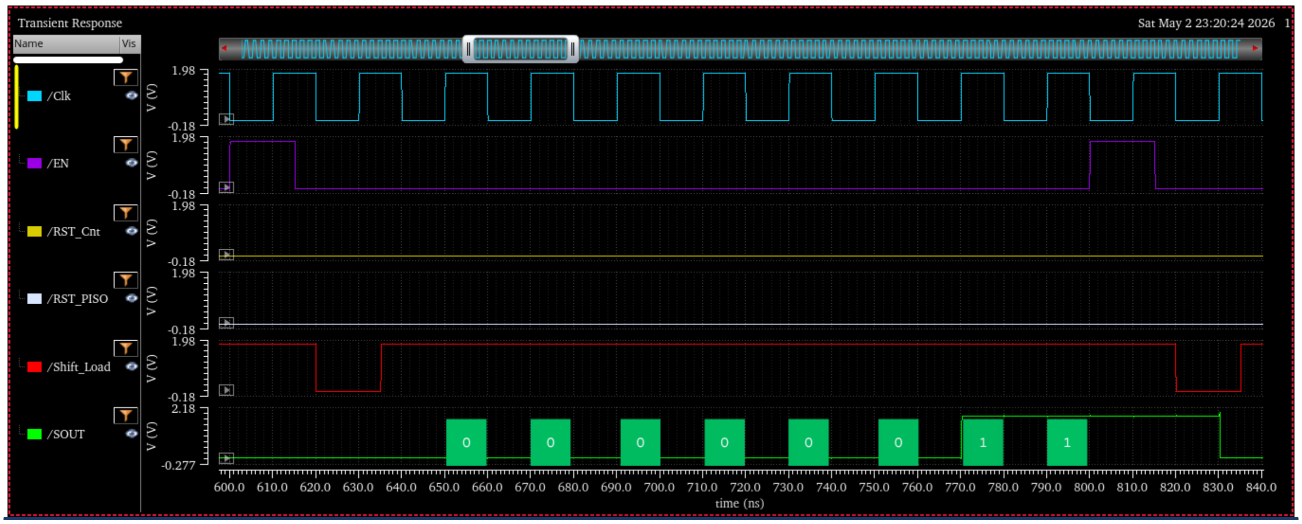


Figure 7: 0x03

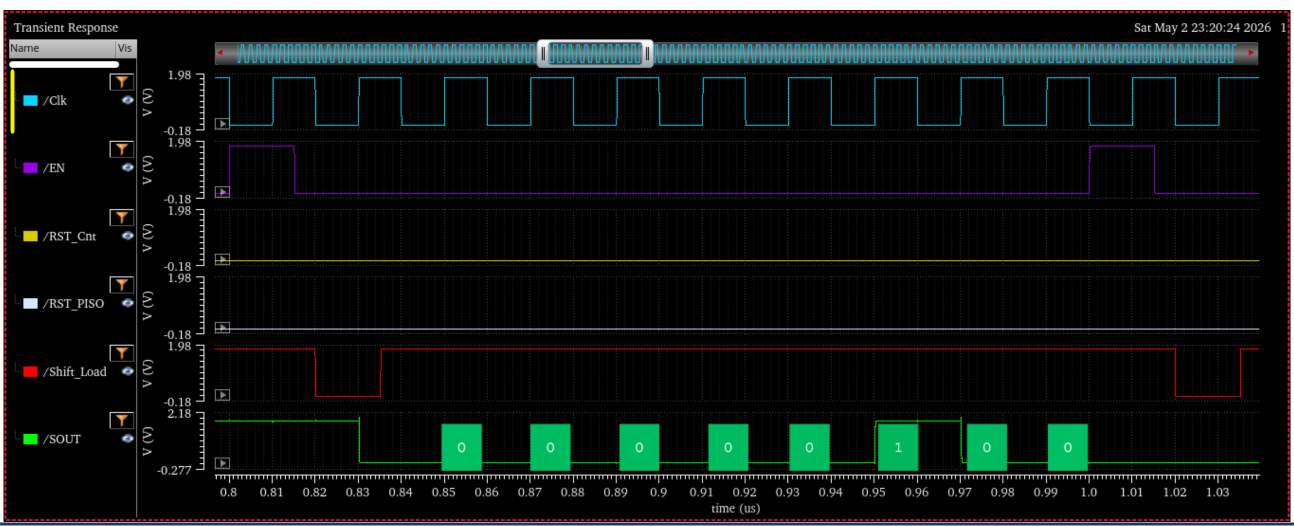


Figure 8: 0x04

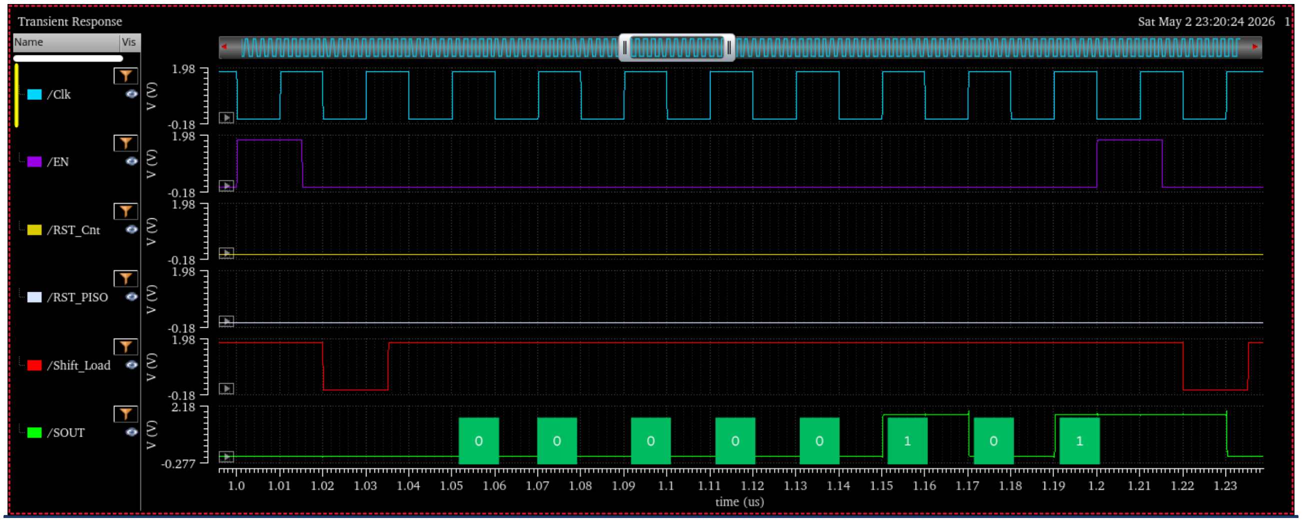


Figure 9: 0x05

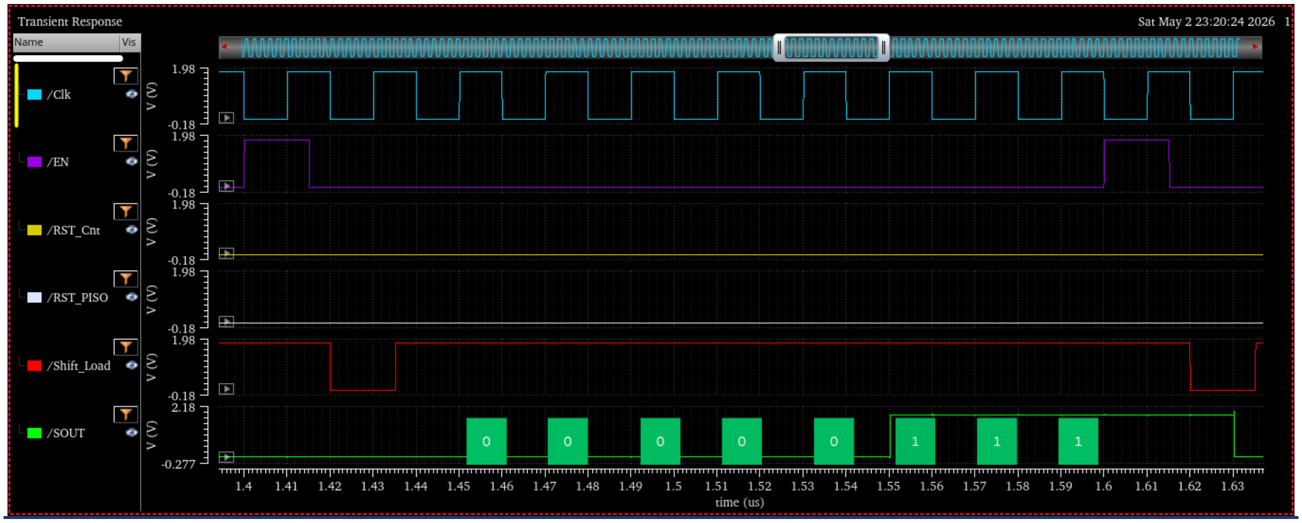


Figure 10: 0x07

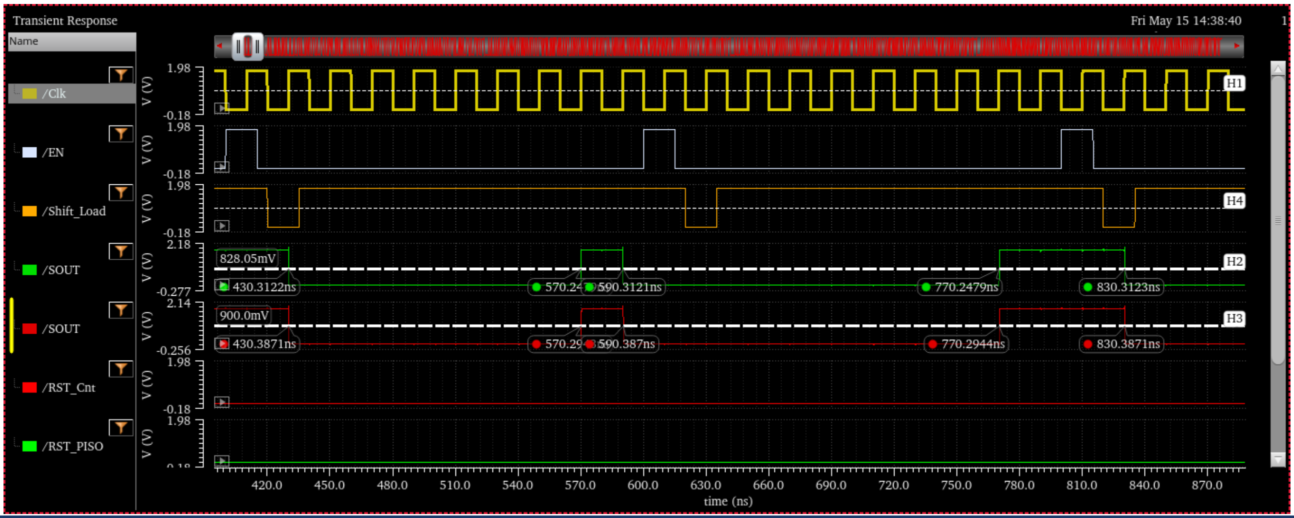
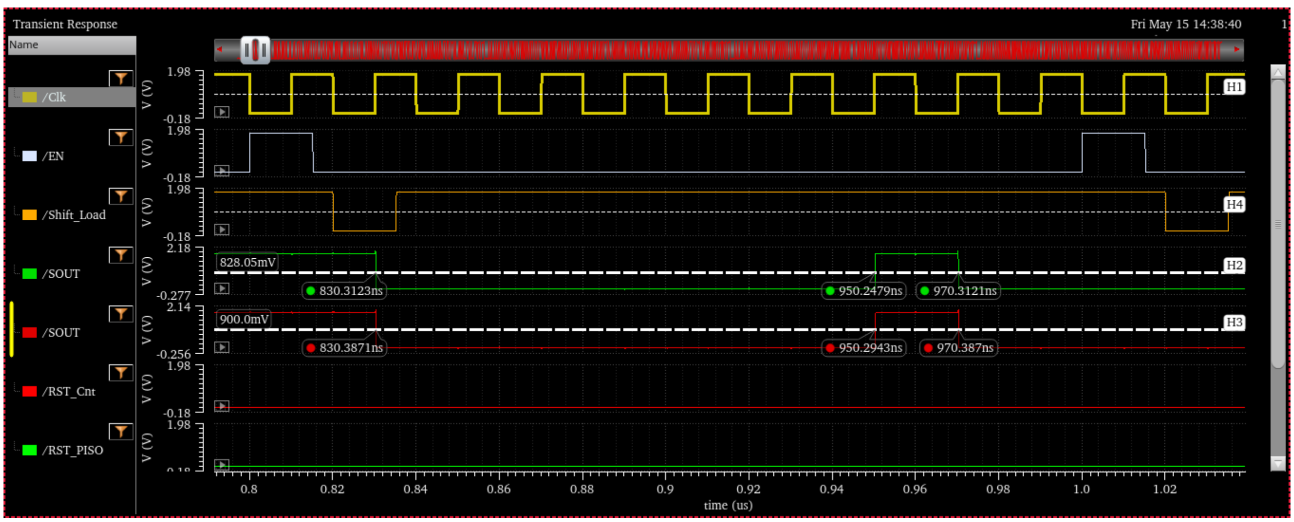


Figure 11: SOUT (Green) - Pre Layout; SOUT (Red) - Post Layout



Parameter	Pre Layout	Post Layout	Delay
T_2	570.2479 ns	570.2943 ns	0.0464 ns
T_3	770.2479 ns	770.2944 ns	0.0465 ns
T_4	950.2479 ns	950.2943 ns	0.0464 ns

Parameter	$t_{50\%}$ SOUT	$t_{50\%}$ Clock	t_{pd} (50% SOUT – 50% CLK)
T_2	570.2479 ns	570.15 ns	0.0979 ns
T_3	770.2479 ns	770.15 ns	0.0979 ns
T_4	950.2479 ns	950.15 ns	0.0979 ns

6 Electrical Characteristics (VDD = 1.8 V, f = 50 MHz)

6.1 Absolute Maximum Ratings

Parameter	Symbol	Value	Unit
Supply voltage	V_{DD}	1.8	V
Input HIGH voltage	V_{IH}	≥ 0.9	V
Input LOW voltage	V_{IL}	≤ 0.9	V
Output HIGH voltage	V_{OH}	1.8	V
Output LOW voltage	V_{OL}	0	V
Clock frequency (tested)	f_{CLK}	50	MHz
Clock period	T_{CLK}	20	ns
Clock pulse width	T_{PW}	10	ns

6.2 Simulation Settings

Setting	Value
Analysis type	Transient
Clock period	20 ns (50 MHz)
Clock pulse width	10 ns
Clock rise / fall	100 ps
Input rise / fall	100 ps
Supply voltage (VDD)	1.8 V
Simulator	Cadence Spectre via ADE-L

7 Current & Power Specifications — 8Bit_Counter_Integrated

Values measured at VDD = 1.8 V, CLK = 50 MHz (period = 20 ns), full count+shift cycle active.

Parameter	Condition	Value	Unit
Average supply current, I_{avg} (Pre-Layout)/(Post-Layout)	CLK @ 50 MHz, all modules switching	**43.82 / 72.35**	µA
I_{max} (Pre-Layout)/(Post-Layout)	SOUT driving logic LOW/HIGH with maximum load capacitance	**5.8169 / 8.7107**	mA
Average power, P_{avg} (Pre-Layout)/(Post-Layout)	$P_{avg} = I_{avg} \times VDD$	**78.89 / 130.23**	µW
Energy per clock cycle (Pre-Layout)/(Post-Layout)	$E = P_{avg} \times 20$ ns	**1.5778/ 2.6046**	pJ
Energy per frame (Pre-Layout)/(Post-Layout)	$E = P_{avg} \times 200$ ns (10 cycles)	**15.778/ 26.046**	pJ
Total Energy consumed (Pre-Layout)/(Post-Layout)	$E = P_{avg} \times t_{sim}$ (t_{sim} = 51200 ns)	**4.039168/ 6.667776**	nJ

8 Module Hierarchy & Dependencies

All cells in both libraries are derived exclusively from the NAND gate. The hierarchy below shows every cell instantiated at each level.

Cell / Module	Function	Library
`8Bit_Counter_Integrated`	Top module	8BitCounter_PHY_ED222 / Gates_ED222_PY
`└ 8Bit_Synchronous_Counter`	Synchronous up-counter	8BitCounter_PHY_ED222
`├ D_FF_with_NAND`	Master-slave D flip-flop	Gates_ED222_PY
`│ ├ NOT_with_NAND`	Inverter (NAND-based)	Gates_ED222_PY
`│ ├ NAND`	2-input NAND gate	Gates_ED222_PY
`│ └ AND_with_NAND`	AND gate (NAND-based)	Gates_ED222_PY
`├ AND_with_NAND`	AND gate (NAND-based)	Gates_ED222_PY
`└ EXOR_with_NAND`	XOR gate (NAND-based)	Gates_ED222_PY
`└ PISO_SR`	Parallel-in serial-out reg.	8BitCounter_PHY_ED222
`├ D_FF_with_NAND`	Master-slave D flip-flop	Gates_ED222_PY
`├ AND_with_NAND`	AND gate (NAND-based)	Gates_ED222_PY
`├ OR_with_NAND`	OR gate (NAND-based)	Gates_ED222_PY
`└ NOT_with_NAND`	Inverter (NAND-based)	Gates_ED222_PY

All 16 D flip-flops in the complete design are the same cell: D_FF_with_NAND.
8 instances reside in 8Bit_Synchronous_Counter | 8 instances reside in PISO_SR.
D_FF_with_NAND is itself built entirely from NAND, AND_with_NAND, and NOT_with_NAND cells.

9 Application Notes

9.1 Receiver-Side Frame Alignment

A receiver reading SOUT must know when each 8-bit frame begins. Assert SHIFT_LOAD = 0 as the frame-start marker: the cycle immediately after EN is the load cycle, and the following 8 cycles carry the valid data. An external 3-bit counter clocked from CLK during SHIFT_LOAD = 1 can track bit position.

9.2 Reading Consecutive Values

To output consecutive integers (0, 1, 2, 3 …) without gaps:

- Cycle 1: Assert EN = 1 — counter increments from N to N+1.
- Cycle 2: Assert SHIFT_LOAD = 0 — PISO captures N+1.
- Cycles 3–10: Assert SHIFT_LOAD = 1 — N+1 is shifted out on SOUT.
- Repeat from step 1. Each frame increments the counter by exactly 1.

9.3 Testbench to test 8-Bit Counter Module

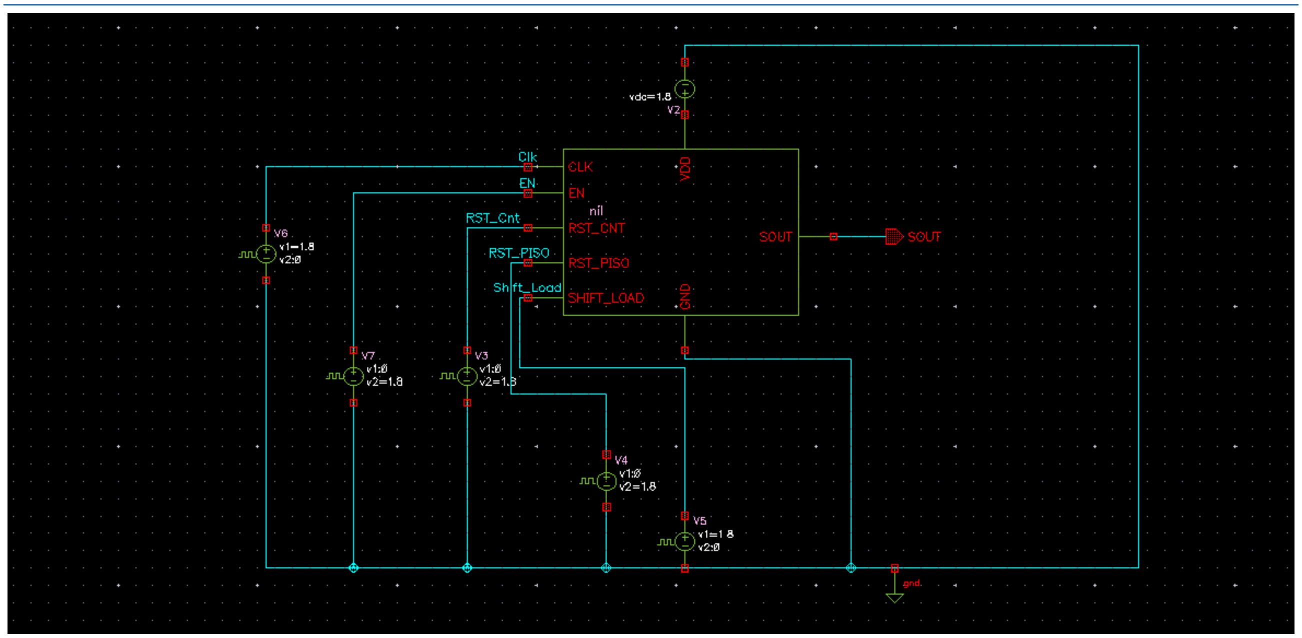


Figure 12: Testbench Schematic for testing 8-Bit Counter

Signal	Voltage 1	Voltage 2	Period	Delay	Pulse width
CLK	1.8 V	0	20 ns	0	10 ns
EN	0	1.8 V	200 ns	200ns	15 ns
RST_CNT	0	1.8 V	100 us	0	15 ns
RST_PISO	0	1.8 V	100 us	0	15 ns
SHIFT_LOAD	1.8 V	0	200 ns	20 ns	15 ns

10 Team Contribution

Task	Contributor(s)
Logic gate schematics (AND, OR, EXOR, DFF, NOT)	All three equally
8-bit synchronous counter schematic	Yash (202404047)
PISO shift register schematic	Prince (202404031)
Testing and waveform verification	Hemin (202404055)
Layout, DRC, LVS & PEX — Logic Gates	Hemin (202404055)
Layout, DRC, LVS & PEX — 8-bit Counter	Prince (202404031)
Layout, DRC, LVS & PEX — PISO	Yash (202404047)
Final system integration	All three equally

Prince (202404031) was responsible for the design and implementation of the PISO shift register schematic, the physical layout of the 8-bit synchronous counter including DRC, LVS, and PEX verification, and participated equally in the design of all logic gate schematics and final system integration.

Yash (202404047) was responsible for the design and implementation of the 8-bit synchronous counter schematic, the physical layout of the PISO shift register including DRC, LVS, and PEX verification, and participated equally in the design of all logic gate schematics and final system integration.

Hemin (202404055) was responsible for testing and waveform verification of all modules, the physical layout of all logic gates including DRC, LVS, and PEX verification, and participated equally in the design of all logic gate schematics and final system integration.